The Summer the Grid Said No

Edition 2 in the Energy, Infrastructure & Climate Resilience series. Previous Decision Intelligence report: 'Powering AI in a Climate-Stressed World' (May 2026).

Cycle: 16 July 2026 · Audience: Senior strategy, foresight, infrastructure, risk and policy professionals · Horizon: 6 to 18 months primary, 2 to 5 years secondary

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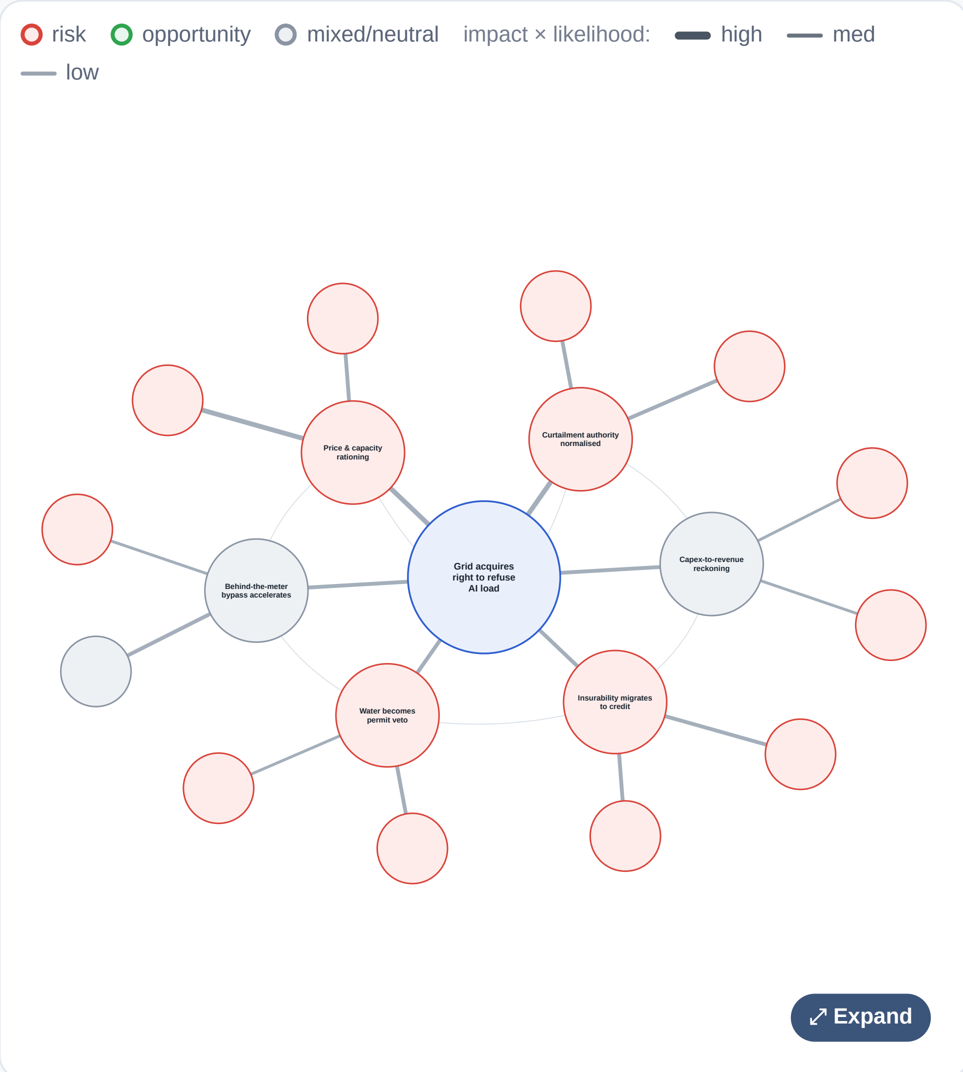
Contents



What you'll learn

1. Why the grid passed its first AI-load heat season by acquiring the right to refuse load, not by building headroom.
2. Why the hyperscaler capex disconfirmation broke upward, and where the risk moved: to returns, not demand.
3. Why softening reinsurance is a misleading signal while climate risk migrates into household credit.
4. Why power-plant water, not on-site cooling, is becoming the binding permit veto on siting.

Key takeaways

1. PJM took two DOE emergency orders to curtail data centres (30 June 2026), warned, then did not need to; demand response absorbed a near-record ~162.7 GW peak.
2. 2026 hyperscaler capex is ~$725bn, up ~77%; every hyperscaler raised guidance; a ~$600bn capex-to-revenue gap is the new fault line.
3. Mid-2026 cat-reinsurance rates fell ~16-20% even as the CA FAIR Plan hit $724bn exposure and the Dallas Fed tied premiums to mortgage delinquency.
4. US data-centre indirect (power) water was ~211bn gallons in 2023, more than 10x on-site cooling; 90.9% of the Southwest was in drought in mid-June 2026.

Executive Synthesis

Why this matters now

The May cycle framed integrated capital planning against four constraints: grid, water, insurance and capital. Eight weeks on, the constraints did not release, they changed venue. The system's first response to a record-demand summer was neither to build nor to price but to acquire the right to refuse load, and five shifts have hardened with it.

Five shifts since the May scan that should be on the board's next agenda:

- **The grid acquired and rehearsed curtailment authority.** Two DOE 202(c) emergency orders let PJM curtail data centres on 30 June; it warned transmission owners, then demand response absorbed a near-record ~162.7 GW peak and physical curtailment "was not required".
- **The capex disconfirmation broke upward.** Every hyperscaler raised guidance, so the buildout thesis strengthened, not the demand-realisation thesis: the capital commitment is confirmed and accelerating and the binding risk migrated from whether the build happens to whether it earns a return.
- **Insurability reversed on price and migrated to credit.** Catastrophe reinsurance became a buyer's market even as the frontier moved into residual markets and household mortgage credit.
- **Water became a permit veto.** Power-plant (indirect) water, not on-site cooling, is the binding ceiling, and it is starting to veto siting.
- **Name the pattern: the Refusal Layer.** Demand that will not bend meets supply that cannot stretch, and a layer decides who gets refused: which load is curtailed, which policyholder non-renewed, which permit vetoed.

This cycle names the constraint that has become the defining variable in AI-infrastructure strategy: not how many megawatts a project needs, but who the grid, the insurer or the water authority is allowed to refuse. Between May and July 2026 that shifted from projection to operating fact. Four developments carry the pattern.

First, the grid passed its first full AI-load heat season, but only by acquiring and rehearsing the authority to refuse data-centre load. PJM obtained two DOE 202(c) orders on 30 June 2026 and rationed by price in parallel, the capacity price clearing at a record $333.44/MW-day and day-ahead power topping $2,000/MWh in parts of PJM. The decisive nuance: on 2 July the grid rehearsed refusal rather than executing it, as demand response absorbed the peak and curtailment "was not required".

Second, the May disconfirmation threshold broke upward. Every hyperscaler raised Q1 guidance; the four are tracking ~$725bn in 2026, up ~77%. The buildout thesis strengthened (the physical build is confirmed and accelerating), so the risk moved to the funding side, where a ~$600bn capex-to-revenue gap now exceeds any prior tech-capex cycle on that measure. The Q2 prints land in late July, after this 16 July publication.

Third, insurability reversed on price while migrating on structure. Global property-cat reinsurance rates fell ~16-20%, the steepest since the late 1990s, yet the frontier did not close: the California FAIR Plan reached $724bn of exposure and the Dallas Fed tied premium rises to mortgage delinquency. The softening cat market is a misleadingly benign signal.

Fourth, water entered the register as a live constraint, absent in May. The dominant burden is indirect: power-generation water was ~211bn gallons in 2023, more than 10x on-site cooling, against a backdrop where 90.9% of the US Southwest was in drought. Water is beginning to act as a permit veto on siting.

The system's first response was neither to build nor to price but to acquire the right to refuse load.

The historical rhyme worth holding is the 2001 telecom bubble: an infrastructure overbuild that was real, useful, and still destroyed the capital that funded it. The question this cycle is not whether the compute gets used but whether its funders are repaid.

Where this analysis could be wrong

The most consequential assumption is that the Refusal Layer is a durable structural shift rather than a market-design artefact. If PJM's Non-Capacity-Backed Load framework is the better reading, curtailment is a deliberate capacity-market instrument, loads trading capacity payments to be first-curtailed, not evidence of physical scarcity. The falsifiable test: if 2027/28 capacity prices ease as queued generation and storage connect, and no RTO ever physically curtails data-centre load through the 2026-27 seasons, then "refusal" was a pricing lever, not a binding ceiling, and the integrated-constraint posture is over-weighted. Signals to watch: physical curtailment actually executed in a subsequent heat event; a named supervisor formalising climate-to-credit transmission into capital rules; a hyperscaler cutting 2026 capex guidance at the late-July Q2 prints.

What changed since the May scan

1. **Grid stress test, graded CONFIRMED via a new mechanism.** May's prediction that summer 2026 would test AI-load adequacy held; the grid coped by acquiring emergency curtailment authority, not by building headroom.
2. **Capex threshold, graded BROKEN UPWARD.** The 15%-cut disconfirmation was not met; every hyperscaler raised. The Q2 prints land late July, after this publication, so an interim check is set for late-July / early-August 2026.

3. **Insurability, graded REVERSED on price, migrating to credit.** May read the frontier as hardening; this cycle it softens on price while migrating on structure into household credit.
4. **Water ceiling, NEW this cycle.** Absent from the May register entirely, it enters as a live constraint with indirect water more than 10x direct cooling.
5. **Proprietary scan delta.** The scan ran 53 sources against May's 41 (including a six-source continuity ledger), holding roughly 57% Tier 1-2, while the evidence base flipped direction on insurance (hardening to softening) and added an entire water theme. The constraints did not release; they changed venue.

Each is developed below, with a decision posture, in the four Strategic Implications.

Continuity ledger: resolving May's dated commitments

May set seven dated obligations. Two (the capex disconfirmation and the summer stress test) are graded in Themes 2 and 1; the remaining five are discharged here, each with a verdict and a post-19-May source, so continuity is resolved explicitly rather than restated.

Obligation	Verdict	What changed since 19 May
PJM colocation rulemaking	UNRESOLVED	The window did not shut, it became a rulemaking: FERC directed PJM to write clear co-location rules (18 December 2025, PJM report due 19 January 2026) and PJM's compliance filing is still in progress.
Firm-clean PPA window	REOPENED	Active on paper, thin on delivered megawatts: nuclear and SMR commitments kept expanding into 2026, but Carnegie (June 2026) judges most hyperscaler pledges largely aspirational against US energy realities this decade.
Hurricane-season posture	CONFIRMED	NOAA (21 May 2026) forecasts a below-normal Atlantic season (55% below-normal; 8-14 named storms, 3-6 hurricanes, 1-3 major) on developing El Nino, easing near-term insured-loss pressure and reinforcing Theme 3's softening read. One landfall would flip it.
CBAM year-one	CONFIRMED	CBAM's definitive phase opened 1 January 2026 alongside an EU Omnibus simplification (a de-minimis exemption removing most small importers); friction was resolved by carve-out, supporting the report's exclusion of CBAM as a 6-18-month energy-cost shock.
Gas-turbine and transformer supply chain	CONFIRMED	The escalation held: data-centre demand drove record GE Vernova orders with turbine slots tight through 2030 and utilities scrambling for transformers, though GE Vernova disputes turbines are the binding gate.

Audience Snapshots

Four functional lenses on the same intelligence base. Each card names the shift and the single question this cycle puts to that function's leadership.

STRATEGY & FORESIGHT

Is your capital matrix still sized around megawatts when the binding variable is refusal?

The shift: the Refusal Layer is now a usable planning frame, demand that won't bend meeting supply that can't stretch; the demand-side uncertainty resolved as capex broke upward; and grid, water, insurance and capital now interact as one integrated-constraint system rather than four separate risk registers.

The question to brief: should we adopt an integrated-constraint capital matrix that scores every FY27 commitment on grid, water, insurance and returns jointly?

RISK, INSURANCE & CFO

Which price signal governs our physical-exposure repricing, the softening cat market or the credit channel?

The shift: catastrophe reinsurance is a buyer's market, a misleadingly benign signal; the Dallas Fed and Bank of England have named the transmission of climate risk into credit; and residual markets like the CA FAIR Plan ($724bn exposure) show where the frontier's discipline actually went.

The question to brief: should we reprice insurance-linked physical exposure through the credit channel now, before the softening cycle lulls underwriting into complacency?

INFRASTRUCTURE, OPERATIONS & CTO

Are we underwriting new sites for water and interconnection timing, or only for megawatts and land?

The shift: interconnection is now curtailable, large loads with backup generation can be ordered to refuse power under DOE 202(c) authority; water has moved from operational line item to permit veto; and municipal systems sized to average cannot absorb data-centre peaking of 3-10x.

The question to brief: should we add a water-availability and interconnection-timing gate to site selection before the next flagship build is committed?

POLICY & PUBLIC AFFAIRS

Is the organisation positioned for the precedents this cycle just set?

The shift: the DOE 202(c) orders establish a live precedent for emergency curtailment of large industrial load; the Bank of England is moving climate risk toward supervisory capital rules; and water-rights disputes such as Utah's Stratos transfer are becoming the decisive local-political gate on siting.

The question to brief: should we engage now on curtailment-terms design, climate-to-credit supervision and water-rights policy, rather than react once the rules bind?

Themes

The cycle's signals are organised into four themes, ranked by impact on near-term decisions. ***Immediate****: changes H2 2026 / H1 2027 capital, siting or offtake decisions.* ***Near-Term****: changes competitive position over the next twelve months.* ***Longer-Range****: a multi-year shift to track each cycle.*

1. The Summer Verdict: the grid passed by acquiring the right to refuse load

IMMEDIATE

The grid passed its first full AI-load heat season, but only after acquiring, and rehearsing, the authority to refuse data-centre load. PJM obtained two DOE 202(c) emergency orders on 30 June 2026, one authorising curtailment of data centres and large loads with backup generation as a last resort, while forecasting a July 2 peak that would break the 2006 record. The refusal was rehearsed, not executed: the actual peak reached ~162,700 MW, suppressed by demand response, and the warned curtailment "was not required". A price channel ran in parallel, with the capacity price hitting a record $333.44/MW-day and an independent monitor attributing 63% of the surge to data centres. Reliability now depends on managing large-load demand at peak, not on headroom.

The summer stress test in four numbers

One record-demand day, four ways the constraint showed up

PJM JULY 2 PEAK	DOE 202(c) ORDERS	CAPACITY PRICE	DAY-AHEAD PEAK
162.7 GW	**2**	**$333**	**>$2k**
near 2006 record 165.6 GW suppressed by demand response	authorised data-centre curtailment, 30 Jun 2026	per MW-day, record (+11x) 63% attributed to data centres	per MWh, parts of PJM roughly 3x last summer
PJM, Jul 2026	PJM / DOE, Jun 2026	Bloomberg, Jul 2026	Bloomberg, Jul 2026

Authorised to refuse load, warned it would — then demand response meant it didn't have to.

The July 2026 PJM heat event in four numbers. Source: PJM Inside Lines, U.S. DOE, Bloomberg.

- PJM obtained two DOE 202(c) emergency orders on 30 June, forecasting a July 2 peak of up to 166,304 MW against the 2006 record of 165,563 MW. PJM Inside Lines (30 June 2026).
- The actual peak reached ~162,700 MW, suppressed by demand response, and the warned data-centre curtailment "was not required and service was not impacted". PJM Inside Lines (3 July 2026).
- Day-ahead power topped $2,000/MWh in parts of PJM, the Western Hub settled near $1,222/MWh (~3x last summer), and the capacity price hit a record $333.44/MW-day, with 63% of the surge attributed to data centres. Bloomberg via Yahoo Finance (3 July 2026).
- The DOE granted authority to direct transmission owners to curtail data centres with backup generation as a last resort before load shed. Utility Dive (2 July 2026); the filing marked an era of managing large-load demand at peak. Maryland Matters (29 June 2026).
- FERC's 2026 Summer Assessment projected demand of 1,587 TWh (+3% YoY), named PJM/NYISO/ISO-NE for the highest prices, and attributed most new load to data centres, the benchmark the May prediction was graded against. US FERC (21 May 2026).
- Europe diverged: ENTSO-E found no systemic adequacy risk, having added 126 GW of renewables and doubled battery storage to 29 GW since the prior summer. ENTSO-E (29 May 2026).
- Clean energy may have been the decisive margin: the US grid set an all-time delivery record, Texas solar covered over 30% of demand at an 83 GW peak, and PJM never needed its emergency diesel generators. Canary Media (10 July 2026).

Counter-argument

The refusal may be a market-design choice, not a reliability collapse. Modo Energy shows PJM's Non-Capacity-Backed Load framework treats curtailment as a deliberate instrument: loads of 50 MW and above forgo capacity payments to be first-curtailed, letting PJM lower its Reliability Requirement and soften auction prices. Grid Strategies reinforces the point, arguing NERC overstates risk because every region held 5-93% more reserves than target once queue resources and imports are counted. The named commercial actors to act on are PJM and peer RTOs as setters of curtailment terms, the demand-response aggregators that absorbed the July 2 peak, and backup / prime-power vendors such as Crusoe (1.6 GW contracted to Meta).

Weak signals to watch

- **WEAK SIGNAL** POWER Magazine reports backup generators converted to prime power, 35+ GW plausibly self-generated by 2030, the same fleet regulators count on for emergency curtailment. Would gain weight if a major campus commissions grid-independent while an RTO cites its backup capacity in a curtailment order.
- **WEAK SIGNAL** An arXiv working paper finds flexible AI load cuts costs 3-21% but does not reduce required capacity, deferral plateauing beyond ~3 hours. Would gain weight if a capacity auction explicitly discounts curtailable large-load flexibility as non-firm.

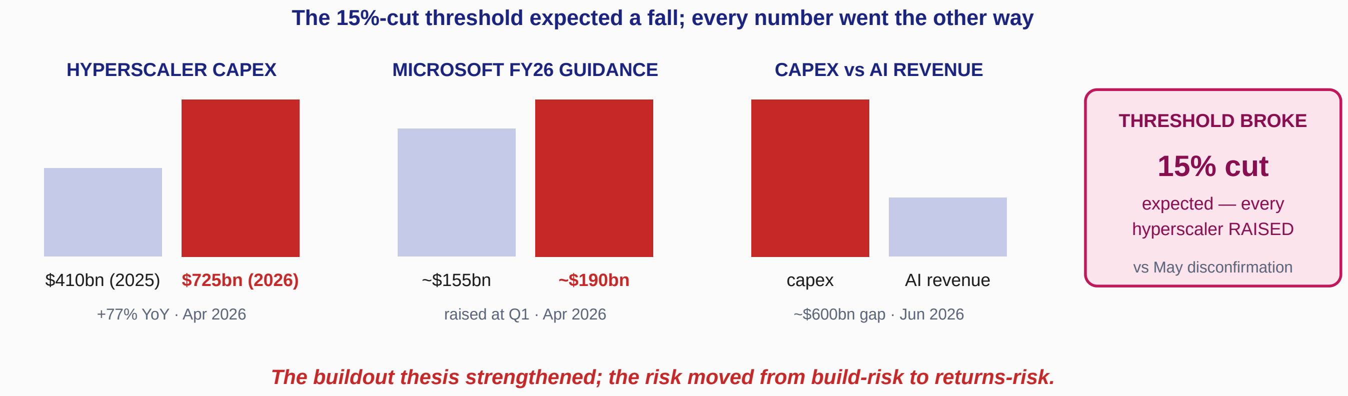
Decision link: Strategic Implications 1.

2. Capex at the Crossroads: the disconfirmation broke upward and the risk moved to returns

IMMEDIATE

The May disconfirming threshold, a 15%+ cut to hyperscaler guidance, was not met. Every disclosing hyperscaler raised 2026 capex at Q1 earnings, so the buildout thesis strengthened, not the demand-realisation thesis: the capital commitment is confirmed and accelerating and the binding risk migrated from whether the build happens to whether it earns a return. The four are tracking ~$725bn in 2026, up ~77% from $410bn in 2025, and the new fault line is a ~$600bn annual gap between AI infrastructure spending and AI-ecosystem revenue, a divergence that exceeds any prior tech-capex cycle on that measure. The build itself is not slowing; the binding constraint has moved upstream to deliverable power. Because Q2 prints land in late July, after this publication, the honest posture is to monitor with a named check date.

The disconfirming threshold broke upward



The capex disconfirmation broke upward at Q1 2026 earnings. Source: Tom's Hardware/Sherwood News, Microsoft IR, Forbes/Allianz.

- Every disclosing hyperscaler raised 2026 capex at Q1: Microsoft to ~$190bn (from ~$155bn), Alphabet to $180-190bn, Meta to $125-145bn. artificialintelligence-news (30 April 2026).
- Meta's Q1 filing reported revenue of $56.3bn (+33% YoY) and raised full-year capex to $125-145bn, citing higher component pricing and data-centre costs. Meta Platforms (29 April 2026).
- The four hyperscalers are tracking ~$725bn in 2026, up 77% from $410bn, with Jefferies calling the bear thesis "garbage". Tom's Hardware (30 April 2026); Sherwood News puts Amazon near $200bn and the total $100bn above prior-quarter expectations (30 April 2026).
- The new fault line is the capex-to-revenue gap: ~$600bn a year, a 46% divergence exceeding the 2001 telecom bubble. Forbes (2 June 2026).
- Allianz quantifies the same tension: capex expanding 46% faster than revenue (vs 32% in 2001) and hyperscaler free cash flow turning negative for the first time in 35 years, while judging the cycle "war-proof for now". Allianz Research (25 March 2026).
- The IEA projects 3.6% average annual global demand growth through 2030 with data centres driving ~half the US increase. IEA Electricity 2026 (26 March 2026); 2025 capex from five firms (>$400bn) is set to rise a further 75% in 2026. IEA (16 April 2026).
- EPRI raised its US data-centre load projection ~60% over 2024, to 9-17% of US electricity by 2030. EPRI (26 February 2026).

- The build is not slowing: SemiAnalysis reports its year-end 2026 self-build forecast has moved only ~1% over six months. SemiAnalysis (18 June 2026); the constraint is now deliverable power, land and permitting. Global Data Center Hub (11 July 2026).

Counter-argument

The demand may be over-attributed and the build over-scaled. IEEFA argues utility data-centre forecasts outstrip independent tech projections by up to a factor of four, risking stranded fossil assets whose costs fall on ratepayers. The cross-domain analogue is the telecom overbuild Allianz benchmarks against: fibre laid in 1999-2001 was genuinely used a decade later, yet the capital that funded it was destroyed. Demand materialising does not guarantee funders are repaid, which is the returns risk this cluster lands on rather than a rival case that the buildout itself is wrong. The named commercial actors to act on are Microsoft, Meta, Alphabet and Amazon as capex setters, SemiAnalysis as the build-rate read, and colocation / power counterparties (Crusoe, Chevron) as the execution layer.

Weak signals to watch

- WEAK SIGNAL David Mytton argues the biggest historical efficiency lever, hyperscale migration, is exhausted and high-end demand forecasts are press-optimised. Would gain weight if a peer-reviewed study forces the IEA/EPRI 2030 ranges downward.
- WEAK SIGNAL Fortune reports expected long-term S&P 500 earnings growth at 20.2%, above 2000's peak, against a ~7% realised average. Would gain weight if AI-exposed multiples compress on a capex-monetisation disappointment at Q2/Q3 prints.

Decision link: Strategic Implications 2.

3. The Insurability Frontier Moves Into Credit

NEAR-TERM

Catastrophe reinsurance softened into a buyer's market, but the insurability frontier did not close, it migrated into residual markets and household mortgage credit. Global property-cat rates fell ~16-20% at mid-2026 renewals, the steepest since the late 1990s, amid abundant capacity and benign losses. Yet the California FAIR Plan reached $724bn of exposure, and central-bank research now ties premium rises to mortgage delinquency and to bank credit losses. The softening cat market sends a misleadingly benign price signal; the discipline moved from price to structure. The binding question for a board is which signal governs physical-exposure repricing.

- Global property-cat rates fell ~16% in 2026, the steepest annual decline since the late 1990s, amid abundant capacity. Guy Carpenter (30 June 2026); mid-year softening was closer to 20% while 2023 terms and conditions largely held. KBW (29 May 2026).
- A $1,000 premium rise pushed ~31,000 mortgages into delinquency in 2022 and is projected to add ~203,000 delinquencies a year between 2025 and 2055, concentrated among low-credit households. Federal Reserve Bank of Dallas (24 March 2026).
- The California FAIR Plan reached $724bn of exposure by December 2025, up 230% since September 2022, having paid ~$3.5bn on Eaton/Palisades claims. CA FAIR Plan oversight hearing (28 January 2026).
- The Bank of England warns climate risks are becoming "more proximate", with rapid asset repricing and UK banks facing climate-related credit losses. Bank of England (25 June 2026).
- June Florida renewals softened further, with double-digit cat-cover declines for some carriers after a hurricane-free 2025 and tort reform. AM Best (20 May 2026).
- Florida's residual market is shrinking even as California's grows: Florida Citizens entered 2026 with 67% less exposure after depopulating 585,432 policies. Florida Citizens data (3 March 2026), falling to a record-low ~278,662 policies. Florida Trend (23 June 2026).
- An NBER working paper finds low-credit-score homeowners pay ~24% more for identical cover, a "second credit channel" rivalling mortgage rates. NBER WP 34848 (1 February 2026).

Counter-argument

The credit-transmission thesis may be premature on pricing grounds. With Gallagher Re confirming a buyer's market and abundant capacity, and Florida's residual market at a decade low, the near-term signal is softening, not stress; the Dallas Fed's ~203,000-delinquency figure is a projection to 2055, not a 6-18-month shock. The counter is that abundant reinsurance capital is masking, not resolving, the physical-loss trajectory, but a board acting only on 2026 pricing would reasonably conclude the frontier is easing. The named commercial actors to act on are Guy Carpenter, Gallagher Re and AM Best as the pricing read, the CA FAIR Plan and Florida Citizens as residual-market gauges, and parametric insurers as the play that captures the risk the standard market sheds.

Weak signals to watch

- **WEAK SIGNAL** The NBER "second credit channel" implies credit score, not just geography, now gates insurability cost. Would gain weight if a rating agency or the GSEs incorporate insurance-credit interaction into mortgage risk models.
- **WEAK SIGNAL** Florida Citizens' record-low count suggests a state can un-harden a frontier through reform and a benign loss year. Would gain weight if California's FAIR Plan exposure begins to fall after comparable reform.

Decision link: Strategic Implications 3.

4. The Water Ceiling: power-plant water as the binding constraint on siting

LONGER-RANGE

The binding water constraint on the AI buildout is not on-site cooling but the power-plant water behind the electricity, and it is beginning to act as a permit veto on siting. Indirect electricity-generation water was ~211bn gallons in 2023, roughly 10-12x direct cooling use, and escaping local scarcity backfires because dry cooling raises electricity demand 25-35%, shifting the burden to thermoelectric plants often themselves in stressed basins. The backdrop is structural scarcity, with 90.9% of the Southwest in drought in mid-June 2026. The empirical case is regional, not national: aggregate thermoelectric water use has fallen, but individual plants can meet or exceed local streamflows, concentrating conflict at the basin scale, where water is repricing project finance.

- Indirect electricity-generation water for US data centres was ~211bn gallons in 2023, roughly 12x the 17.4bn gallons of direct cooling, though ITIF argues it is a soluble governance problem. ITIF (6 July 2026).
- Dry cooling raises electricity demand 25-35%, shifting the water burden to power plants; per-community burden spans 0.2-134% of host peak-day capacity. arXiv (WINGS/UC Riverside) (19 June 2026).
- 90.9% of the Southwest US was in drought as of mid-June 2026, with monsoon rains having little influence on Colorado/Rio Grande streamflow. NOAA/NIDIS (18 June 2026).
- Data-centre-driven power-generation water use is projected to surge ~400% (2.9 to 14.5bn gal/yr), raising stress in strained basins by up to 17% annually. Ceres (23 September 2025).
- Aggregate US thermoelectric water use fell 2008-2020, yet some plants withdraw volumes meeting or exceeding local streamflows, concentrating conflict at the basin scale. USGS (20 September 2025).
- ~30% of capacity under construction (~$550bn completion value) sits where water scarcity is expected to intensify, naming Amazon, Alphabet, Meta and Microsoft. MSCI (9 December 2025).
- Water is becoming a permit veto in practice: Utah's Stratos developers withdrew a 1,900 acre-feet transfer after ~3,900 protests. Utah News Dispatch (8 May 2026); a Georgia authority told a 6-million-gallon/day project "we just don't have the water". Data Center Knowledge (29 May 2026).
- Water has shifted from operational line item to permit veto, a mid-construction refusal forcing a 12+ month delay and covenant stress. Global Data Center Hub (15 May 2026).

Counter-argument

Water may not be the binding gate at all, interconnection is. Works in Progress argues the primary bottleneck is grid interconnection, with median wait times rising from under 20 months in 2005 to 55 months in 2023 and 62% of data centres considering off-grid solutions because of queue delays, not water. S&P Global notes water is under 1% of operating cost, so near-term financial materiality is constrained; the risk is regulatory and reputational, not yet economic. The named commercial actors to act on are S&P Global, MSCI and Ceres as the risk-raters, siting-advisory and closed-loop / dry-cooling vendors, and water-rights counterparties.

Weak signals to watch

- WEAK SIGNAL The WINGS/UCR feedback-loop paper shows per-community water burden spanning 0.2-134% of host peak-day capacity, i.e. the constraint is local and peak, not aggregate. Would gain weight if a second peer-reviewed study replicates the burden-shift finding at additional sites.
- WEAK SIGNAL The "permit veto" underwriting lens suggests lenders are beginning to price water-withdrawal refusal into covenants. Would gain weight if a rated project-finance deal is publicly repriced on a water-permit contingency.

Decision link: Strategic Implications 4.

Strategic Implications

Four decisions the cycle brings forward, each with an owner, a dated action and a decision posture (Decide, Prepare, Monitor). One opportunity line names who profits from each constraint.

SI 1: Take a capital-committed position in the Refusal Layer's flexibility tools

The July episode proved that the value accrued not to the megawatts curtailed (none were) but to the flexibility that absorbed the peak. Treat curtailment authority as the new operating baseline for large-load interconnection, not a transient episode, and build a documented position in demand-response and grid-flexibility capacity now, ahead of the next heat event. If curtailment proves a capacity-market artefact on the Modo reading, flexibility premia compress, so the commitment should be staged and hedged against that falsification.

Action: by 30 September 2026, table a capital-committed position paper on demand-response / grid-flexibility capacity for FY27 sign-off, ahead of the next heat event.

Who profits: demand-response and flexibility aggregators, transformer and backup-power supply-chain positions, and closed-loop cooling, the assets the grid is effectively buying an option on.

DECIDE

Draws on Theme 1. Owner: CSO / CFO.

SI 2: Re-grade the capex thesis at the Q2 prints and hold the integrated-constraint matrix

The trade-off has inverted: stop hedging against a demand shortfall and start hedging against a returns shortfall, as a ~$600bn capex-to-revenue gap now exceeds any prior tech-capex cycle on that measure. Because Q2 2026 prints land in late July, after this publication, the honest posture is to monitor with a named check date, holding the integrated-constraint matrix carried from May and re-grading against the first hard read. The SemiAnalysis build-rate read and the Allianz free-cash-flow trajectory are the indicators that would flip this from Monitor to Decide before October.

Action: by early August 2026 (post Q2 prints), re-grade the capex thesis and integrated-constraint matrix; escalate if two or more hyperscalers cut 2026 guidance.

Who profits: those in the execution layer, deliverable-power, colocation and firm-clean supply counterparties (Crusoe, Chevron), as the binding constraint moves upstream from capital to power delivery.

MONITOR

Draws on Theme 2. Owner: CSO.

SI 3: Reprice climate exposure through the credit channel, not the softening cat-cover channel

The forced choice is which price signal to believe: the softening cat-cover market says "relax", the credit channel says "reprice". Reprice insurance-linked physical exposure through the credit channel the Dallas Fed and Bank of England have named, not through the softening cat-cover channel sending a misleadingly benign signal, acting before the softening cycle lulls underwriting into complacency. The capability required is parametric structuring, basis-risk modelling and the "second credit channel" analytics. If losses stay benign the repricing over-provisions, so the move is a repricing of exposure and cost of capital, not a withdrawal of capacity.

Action: by 30 September 2026, reprice insurance-linked physical exposure through the credit channel in the FY27 plan and cost of capital, not the cat-cover channel.

Who profits: parametric insurers and structured-risk providers that price the exposure the standard market is refusing, the venue where refusal, not price, is doing the work.

DECIDE

Draws on Theme 3. Owner: CFO / CRO.

SI 4: Add a water-availability and interconnection-timing gate to siting and project-finance underwriting

The constraint is real but regional and permit-mediated, so the forced choice is siting discipline, not a blanket build freeze. Introduce a water-availability and interconnection-timing gate into siting due diligence and project-finance underwriting before the next flagship build is committed, treating a permit veto as a modelled downside. The capability required is basin-level water-rights analytics of the kind Ceres and MSCI supply, interconnection-queue modelling and permitting advisory. If ITIF is right that water is a soluble governance problem, the gate under-monetises, which is why the posture is Prepare rather than Decide.

Action: by the October 2026 cycle, embed a water-availability and interconnection-timing gate into siting due diligence and project-finance underwriting for all >500 MW candidates.

Who profits: water-siting advisory, basin-stress analytics and closed-loop / dry-cooling vendors that sell directly into the siting gate.

PREPARE

Draws on Theme 4. Owner: COO / General Counsel.

Scenario Matrix

Four 18-month operating environments generated by crossing two axes the evidence base does not yet decide: whether AI capital discipline holds or breaks, and whether the physical/infrastructure constraint stays manageable or binds hard across grid, water and climate together. The matrix is a planning tool, not a forecast; the value sits in the indicators that would tip a reader from one cell to another. The axes evolved from May's ("AI capex sustained vs corrected" × "climate loss accelerates vs moderates"): the demand-side uncertainty resolved as capex broke upward, so the X-axis moves to the funding-side question, and the climate axis broadens from loss-acceleration to whether the physical constraint binds hard.

AI CAPITAL DISCIPLINE

DISCIPLINE HOLDS

DISCIPLINE BREAKS

PHYSICAL/INFRASTRUCTURE CONSTRAINT

The Refusal Economy

The worst pairing: capital discipline breaks and the physical constraint binds. A subsequent heat event forces actual curtailment, water vetoes multiply, the capex-to-revenue gap reprices, and the Bank of England's "more proximate" climate-credit risk crystallises. Refusal is enforced across every venue at once while the capital funding the build retreats mid-cycle, stranding partially-built infrastructure in constrained basins.

Indicators: physical curtailment executed and a hyperscaler capex cut in the same window; climate-credit transmission formalised into supervisory capital rules; project-finance deals restructured on both power and water contingencies.

Build or Be Refused

Capital stays disciplined and demand is real, but the physical layer cannot keep pace: a heat event forces actual curtailment of data-centre load, water permits veto flagship builds, and Southwest drought tightens siting. The Refusal Layer moves from rehearsal to enforcement across grid, water and insurance. Growth continues but is rationed, and whoever holds flexibility, firm power and water rights sets the terms.

Indicators: an RTO physically curtails data-centre load in a 2026-27 heat event; a second named water-permit veto halts a >500 MW campus; capacity prices stay at or above the cap into 2028.

The Capex Hangover

The physical system copes, but the funding side does not: the ~$600bn capex-to-revenue gap forces a repricing, hyperscaler free cash flow strain bites, and a bubble-style correction, the 1999 rhyme, slows the build from the capital side. Grid and water constraints ease as demand growth moderates, but stranded-asset risk on committed generation rises. The compute gets built and used, but its funders are not repaid.

Indicators: a hyperscaler cuts 2026 capex guidance materially at Q2/Q3 prints; visible equity repricing of AI-exposed "asset-light" multiples; self-build forecasts turn down more than 5%.

CONSTRAINT MANAGEABLE

Managed Glidepath

Hyperscalers hold capex discipline as the gap narrows through monetisation, while queued generation and storage connect fast enough that curtailment authority stays a rehearsed backstop. Reinsurance stays a buyer's market, water stays a soluble governance problem, and the Refusal Layer remains a set of authorities held but not used. Constraints are managed by pricing and permitting, never enforced.

Indicators: Q2/Q3 hyperscaler capex guidance flat-to-down without demand deterioration; 2027/28 PJM capacity price falls back below the cap as queue resources connect; no physical data-centre curtailment executed through t

CONSTRAINT BINDS HARD

What We Are Not Planning For

Four scenarios held out of the plan because the evidence base does not yet justify resourcing against them. Each carries the reinstatement trigger that would change that judgement.

Exclusion 1: Routine physical curtailment of data-centre load as a base case

This cycle demand response absorbed a near-record peak and curtailment was warned but "not required". We are not planning for flexibility to fail so completely that physical curtailment becomes a routine summer operating tool rather than a rehearsed backstop; the July evidence points the other way.

Reinstatement trigger: *an RTO physically curtails data-centre load in a 2026-27 heat event, or a capacity auction discounts curtailable large-load flexibility as non-firm.*

Exclusion 2: A near-term collapse in US federal climate and energy policy

The constraints this cycle names operate through grid operators, insurers and water authorities, venues that bind largely independent of the federal climate-policy direction. We are not planning for a sudden federal reversal, or a decisive federal pre-emption of state siting, water and supervisory action, to reshape the 6-18-month picture; the binding action is at RTO, state and central-bank level.

Reinstatement trigger: *a federal action materially pre-empting state water/siting authority or removing the climate-to-credit supervisory direction of travel within the planning horizon.*

Exclusion 3: CBAM as an immediate energy-cost shock (carried forward from May)

May deprioritised the carbon border adjustment mechanism as an immediate shock; nothing this cycle changes that. The dominant near-term energy-cost pressures are grid capacity pricing and interconnection, not carbon-border adjustment. We continue to treat CBAM as a structural 3-5-year factor, not a 6-18-month planning input.

Reinstatement trigger: *a finalised CBAM schedule with a dated, material pass-through to electricity or embedded-carbon input costs within the planning horizon.*

Exclusion 4: Off-grid AI campuses as the dominant model (carried forward and updated from May)

May excluded off-grid campuses as the dominant model; we maintain that, with a sharper watch. Prime-power self-generation is scaling (35 GW plausible by 2030), now a live weak signal rather than a dismissed one, but grid-connected remains the majority model, so we do not plan around off-grid as the base case.

Reinstatement trigger: *a flagship >500 MW campus commissioning grid-independent, or self-generation share crossing a materiality threshold in RTO planning documents.*

Discussion Points

1. If the grid can pass a record-demand day only by holding curtailment authority in reserve, do we treat that authority as a transient episode or reprice all large-load interconnection exposure around it this cycle?
2. **For investors:** if the capex-to-revenue gap already exceeds any prior tech-capex cycle on Allianz's measure and Q2 prints land after this publication, do we hedge against a returns shortfall rather than the demand shortfall planned for in May, and what guidance movement at late-July earnings forces us to reconvene before October?
3. If reinsurance is softening while the Dallas Fed projects ~203,000 additional annual mortgage delinquencies, which price signal governs our physical-exposure repricing, the softening cat market or the credit channel the Bank of England is now naming?
4. If backup generators are becoming prime power that lets loads bypass the grid entirely, is the curtailment authority the grid just acquired already depreciating, and does our flexibility thesis survive that?
5. If power-plant water is more than 10x direct cooling and dry cooling shifts the burden to stressed basins, do we introduce a basin-stress and interconnection-timing gate into siting before, not after, the next flagship commitment?

Source Confidence Register

53 verified sources across four themes plus a five-verdict continuity ledger. Tier 1 (governments, regulators, multilateral bodies, national statistics, primary filings): 17. Tier 2 (institutional research, think-tanks, consultancies, reinsurer catastrophe reports): 13. Tier 3 (quality journalism and specialist trade press): 23. No Tier 4 vendor/advocacy sources are load-bearing. Non-English originals: 0. Recency window 0-12 months, most sources post-19-May 2026; ten sources 6-12 months old are flagged *(potentially dated)* and used for structural context, not as a fresh board-level anchor. 30 of the 53 sources (57%) sit at Tier 1 or 2; the source set was frozen at 2026-07-14. Every claim was checked against genuinely fetched page text in the source-verification pass. Detect → Assess → Decide → Act.

Theme 1: The Summer Verdict, grid performance under the first full AI-load heat season

Source	Tier	Date	Key claim
Utility Dive	T3	2 Jul 2026	DOE granted authority to direct transmission owners to curtail data centres with backup as a last resort before load shed.
PJM Inside Lines (30 Jun)	T1	30 Jun 2026	Two DOE 202(c) emergency orders; July 2 peak forecast up to 166,304 MW vs the 2006 record of 165,563 MW.
PJM Inside Lines (3 Jul)	T1	3 Jul 2026	Actual peak ~162,700 MW, suppressed by demand response; warned curtailment "not required and service was not impacted".
Maryland Matters	T3	29 Jun 2026	PJM DOE filing marks an era where reliability depends on managing large-load demand at peak stress.
Bloomberg via Yahoo Finance	T3	3 Jul 2026	Day-ahead >$2,000/MWh in parts of PJM; capacity price a record $333.44/MW-day; 63% of the surge attributed to data centres.
US FERC 2026 Summer Assessment	T1	21 May 2026	Summer demand 1,587 TWh (+3% YoY); PJM/NYISO/ISO-NE highest prices; most new load attributed to data centres.
ENTSO-E 2026 Summer Outlook	T1	29 May 2026	No systemic European adequacy risk; 126 GW renewables added and battery storage doubled to 29 GW.
Grid Strategies	T2	6 Jul 2026	NERC overstates risk; every region held 5-93% more reserves than target once queue resources and imports are counted.
Canary Media	T3	10 Jul 2026	US grid set an all-time delivery record; Texas solar >30% of demand; PJM never needed its emergency generators.

Source	Tier	Date	Key claim
Modo Energy	T2	5 Sep 2025	PJM Non-Capacity-Backed Load framework treats curtailment as a market-design instrument, not physical failure. *(potentially dated)*
arXiv (To Defer or To Shift?)	T1	8 Apr 2026	Flexible AI load cuts costs 3-21% but does not reduce required capacity; deferral plateaus beyond ~3 hours.
POWER Magazine	T3	18 May 2026	Backup generators converted to prime power; 35+ GW self-generated by 2030, 27% of data centres onsite-only.

Theme 2: Capex at the Crossroads, hyperscaler guidance vs the disconfirming threshold

Source	Tier	Date	Key claim
IEA Electricity 2026	T1	26 Mar 2026	3.6% average annual global demand growth 2026-30; data centres drive ~half the US increase.
IEA (data-centre use surged)	T1	16 Apr 2026	Data-centre demand rose 17% in 2025; >$400bn 2025 capex from five firms set to rise 75% in 2026; doubling by 2030.
Meta Platforms Q1 2026	T1	29 Apr 2026	Q1 2026 revenue $56.3bn (+33% YoY); full-year capex raised to $125-145bn.
artificialintelligence-news	T3	30 Apr 2026	All hyperscalers raised 2026 capex: Microsoft ~$190bn, Alphabet $180-190bn, Meta $125-145bn.
Tom's Hardware	T3	30 Apr 2026	Four hyperscalers on track for $725bn in 2026, up 77%; Jefferies calls the bear thesis "garbage".
Sherwood News	T3	30 Apr 2026	Big four plan $700bn+ 2026 capex (Amazon ~$200bn), $100bn above prior-quarter expectations.
EPRI Powering Intelligence 2026	T2	26 Feb 2026	US data-centre load projection raised ~60% over 2024; 9-17% of US electricity by 2030.
Allianz Research	T2	25 Mar 2026	Capex expanding 46% faster than revenue (vs 32% in 2001); hyperscaler free cash flow negative first time in 35 years.
Forbes	T3	2 Jun 2026	~$600bn annual capex-to-revenue gap, a 46% divergence exceeding the 2001 telecom bubble.
IEEFA	T3	3 Jun 2025	Utility data-centre forecasts outstrip independent tech projections up to 4x; stranded-asset risk. *(potentially dated)*

Source	Tier	Date	Key claim
dev/sustainability (Mytton)	T3	11 May 2026	Hyperscale efficiency lever exhausted; extreme high-end demand forecasts are press-optimised.
SemiAnalysis	T3	18 Jun 2026	Year-end 2026 self-build forecast moved only ~1% in six months; "50% cancelled" narrative overstated.
Global Data Center Hub (roundup)	T3	11 Jul 2026	Binding constraint has shifted upstream to deliverable power, land and permitting (Crusoe, ASEAN, Chevron deals).
Fortune	T3	8 Jun 2026	Expected long-term S&P 500 earnings growth 20.2%, above 2000's peak, vs ~7% realised average.

Theme 3: The Insurability Frontier Moves Into Credit

Source	Tier	Date	Key claim
Reinsurance News / KBW	T3	29 May 2026	Mid-year property-cat renewals softened ~20%; 2023 terms and conditions largely held.
AM Best (via Artemis)	T2	20 May 2026	June 2026 Florida renewals softening more; double-digit cat-cover declines for some carriers.
Guy Carpenter (via Artemis)	T2	30 Jun 2026	Global property-cat rates down ~16% in 2026, steepest fall since the late 1990s.
Gallagher Re (via Artemis)	T3	13 Jul 2026	Cat rates down up to 20% for loss-free accounts; incumbent capacity adequate, a clear buyer's market.
CA FAIR Plan oversight hearing	T1	28 Jan 2026	California FAIR Plan exposure $724bn by Dec 2025, up 230% since Sept 2022; ~$3.5bn Eaton/Palisades claims paid. *(potentially dated)*
Florida Citizens (via Artemis)	T2	3 Mar 2026	Florida Citizens entered 2026 with 67% less exposure after depopulating 585,432 policies.
Florida Trend	T3	23 Jun 2026	Florida Citizens at a record-low ~278,662 policies; commissioner calls the market healthiest in a decade.
Federal Reserve Bank of Dallas	T1	24 Mar 2026	$1,000 premium rise pushed ~31,000 mortgages into delinquency (2022); ~203,000/yr projected 2025-55.
NBER WP 34848	T1	1 Feb 2026	Low-credit-score homeowners pay ~24% more for identical cover, a "second credit channel".
Bank of England	T1	25 Jun 2026	Climate risks "more proximate", materialising sooner; rapid repricing and UK banks facing climate credit

Source	Tier	Date	Key claim
			losses.

Theme 4: The Water Ceiling, power-plant water as the binding constraint on siting

Source	Tier	Date	Key claim
arXiv (WINGS / UC Riverside)	T1	19 Jun 2026	Dry cooling raises electricity demand 25-35%, shifting water burden to power plants; burden 0.2-134% of host capacity.
ITIF	T2	6 Jul 2026	Indirect power-plant water ~211bn gallons in 2023, ~12x direct cooling, but soluble via watershed standards.
Ceres	T2	23 Sep 2025	Power-generation water use projected to surge ~400% (2.9 to 14.5bn gal/yr); basin stress up to +17% annually. *(potentially dated)*
NOAA / NIDIS	T1	18 Jun 2026	90.9% of the Southwest US in drought mid-June 2026; monsoon little influence on Colorado/Rio Grande streamflow.
USGS	T1	20 Sep 2025	Aggregate thermoelectric water use fell 2008-2020, yet some plants exceed local streamflows; conflict is basin-scale. *(potentially dated)*
MSCI	T2	9 Dec 2025	~30% of capacity under construction (~$550bn) sits where water scarcity is expected to intensify. *(potentially dated)*
S&P Global Ratings	T2	15 Sep 2025	43% of data centres in high-water-stress areas, but water <1% of opex; risk is regulatory, not yet economic. *(potentially dated)*
Works in Progress	T3	22 Jun 2026	Primary bottleneck is interconnection (waits 20 to 55 months), not water; 62% consider off-grid.
Global Data Center Hub (water)	T3	15 May 2026	Water shifted from operational line item to "permit veto"; mid-construction refusal forces a 12+ month delay.
Utah News Dispatch	T3	8 May 2026	Stratos developers withdrew a 1,900 acre-feet water transfer after ~3,900 protests.
Data Center Knowledge	T3	29 May 2026	Georgia authority: "we just don't have the water"; municipal systems can't absorb 3-10x data-centre peaking.

Continuity Ledger — resolving May's dated commitments

Source	Tier	Date	Key claim
U.S. Federal Energy Regulatory Commission	T1	18 Dec 2025	FERC directed PJM to write clear co-location rules (report due 19 Jan 2026). *(potentially dated)*
Utility Dive	T3	19 Dec 2025	Window converted into a tariff rulemaking, still in progress. *(potentially dated)*
Carnegie Endowment for International Peace	T2	Jun 2026	Hyperscaler nuclear pledges largely aspirational vs US energy realities this decade.
NOAA	T1	21 May 2026	Below-normal 2026 Atlantic season (55%); 8-14 named storms, 3-6 hurricanes, 1-3 major.
International Carbon Action Partnership	T2	Jan 2026	CBAM definitive phase live 1 Jan 2026 with Omnibus de-minimis simplification. *(potentially dated)*
Power Engineering	T3	Jun 2026	Record GE Vernova orders; turbine slots tight through 2030; transformer scramble.

Conflict notes: Three genuine source disagreements were weighted rather than resolved. (1) Grid reliability: NERC's "elevated risk" labels vs Grid Strategies' "overstated" rebuttal and Modo's "curtailment is market design, not failure" reading; the report weights the observed rehearsed-not-executed outcome and names the market-design framing as the live falsification test. (2) Capex: the IEA/EPRI demand base case vs IEEFA's overbuild critique and Mytton's efficiency scepticism, weighted toward the base case for the 6-18m horizon while carrying the overbuild/returns critique as the dominant fault line. (3) Water: the "binding ceiling" thesis vs Works in Progress (interconnection is the real gate) and ITIF (soluble governance); the report frames water as a regional, permit-mediated gate, not a national ceiling. Author-type coverage: the set is strong on analysts, policy advisors, scientists and economists but thin on innovators, change agents and consultants (only Guy Carpenter and Gallagher Re represent the consulting lens), and business-leader voice rests on a single primary filing (Meta); treat vendor/operator claims (POWER Magazine, Global Data Center Hub) as directional.

Claim-Fidelity Appendix: Analyst inferences and editorial framing

This briefing carries a set of analyst-generated interpretations that go beyond what any single source asserts. They are named here so a reader can trace the confidence line and disagree productively.

Analytical inferences carried by the body prose

- "The Refusal Layer" is an analyst frame linking four independently sourced constraints (grid curtailment, insurance non-renewal, water permitting, capital discipline) into one pattern; no source uses the term.
- Grading the May capex disconfirmation as "broken upward" is our interpretation: the sources report the guidance raises; the judgement that this strengthens the thesis and moves the risk to returns is ours.
- Treating softening reinsurance as a misleading signal is an inference: the broker reports establish the softening as fact; characterising it as a benign signal that masks credit-channel migration is our synthesis of the Dallas Fed, Bank of England and FAIR Plan evidence.
- Casting power-plant water as a system ceiling rather than a siting constraint is a framing choice contested by ITIF and S&P Global; we adopt the "regional, permit-mediated gate" reading, itself an analyst reconciliation of conflicting sources.
- The 2001-telecom analogy and the "rehearsed, not executed" reading of the July episode are interpretive overlays on the PJM operational notices, not claims those notices make.

Editorial framing

The title "The Summer the Grid Said No" is an editorial compression: the grid's refusal of data-centre load was authorised and rehearsed (two DOE 202(c) orders, transmission owners warned) but not physically executed, as demand response absorbed the peak. Consolidating four parallel constraints under the single "Refusal Layer" heading is likewise an editorial choice that compresses independent analyses into one decision-actionable synthesis; readers who prefer to treat the four themes as separate risk registers can do so without losing any underlying claim.